

Patient Appointment No-Show Prediction

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ISOM 835 – Predictive Analytics and Machine Learning

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Introduction & Dataset Description

The Patient Appointment No-Show Prediction Dataset focuses on a realistic healthcare Dataset for Predicting Missed Appointments Using ML. Linked here: <https://www.kaggle.com/datasets/mirzayasirabdullah07/patient-appointment-no-show-prediction-dataset>. This dataset simulates real-world patient appointment behavior, focusing on predicting no-shows in healthcare systems. "No-shows in healthcare are scheduled appointments that patients miss without providing prior notification to the clinic or provider. These incidents, often termed "did not attend" (DNAs), create significant disruptions in care, wasting staff time and resources, while causing lost revenue and worsening patient health outcomes (Gasida, 2026). It includes key factors such as patient demographics, booking patterns, waiting time, reminders, and past attendance history. The dataset is designed for building classification models, analyzing patient behavior, and developing AI-driven solutions to reduce missed appointments and improve healthcare efficiency.

The dataset head:

patient_no_show_dataset.csv (118.8 KB)



Detail Compact Column

8 of 8 columns

About this file

This dataset simulates real-world patient appointment behavior, focusing on predicting no-shows in healthcare systems. It includes key factors such as patient demographics, booking patterns, waiting time, reminders, and past attendance history. The dataset is designed for building classification models, analyzing patient behavior, and developing AI-driven solutions to reduce missed appointments and improve healthcare efficiency.

📅 booking_date	📅 appointment_date	# age	⚧ gender	# waiting_days	# sms_re
Date when the patient scheduled the appointment. This helps analyze booking behavior and lead time	The actual date of the scheduled appointment. Used to understand attendance patterns over time.	Age of the patient in years. Can be used to analyze how different age groups impact appointment attendance.	Gender of the patient (Male/Female). Useful for demographic-based analysis.	Number of days between booking_date and appointment_date. Higher waiting times often increase the	Indicates patient reminder
			Female 50% Male 50%		
2023-12-05 2024-12-29	2023-12-31 2024-12-29	1 89		0 29	0
2024-03-24	2024-04-12	15	Male	19	1
2024-07-23	2024-08-02	75	Female	10	0
2024-11-03	2024-11-04	88	Female	1	1
2024-05-29	2024-06-09	58	Female	11	0
2024-05-22	2024-06-18	60	Female	27	0
2024-12-07	2024-12-20	64	Male	23	0

Dataset Selection Rationale

I initially thought about looking at a healthcare dataset after my coworker had mentioned she had been put on a six-month waiting list to be seen for her acid reflux. It's keeping her up at night and affecting her day-to-day. She had gone to a specialist, but he said he could not help and that she should go to a gastrologist instead. It made me wonder how often people just miss these appointments that someone like her would be dying to get. How large an issue is this? My personal interest had me do a little research. That's when I found a data set that actually explored how

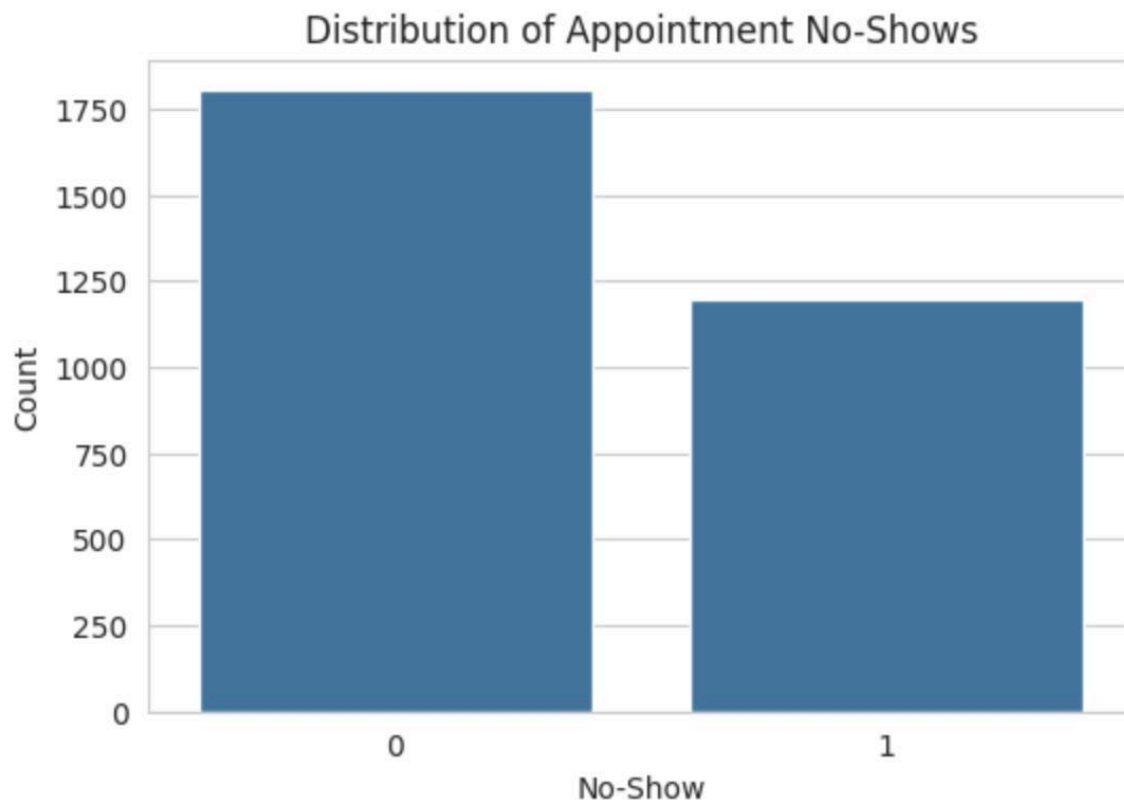
Patient no-shows are a massive, high-impact issue in healthcare, costing the US system over \$150 billion annually and leading to 14% daily revenue losses per practice. This [Kaggle](#)

Dataset addresses a critical challenge where 25% or more of appointments are missed, disrupting care continuity and wasting scarce resources.”

Patient Appointment No-Show Prediction Dataset focuses on a Realistic Healthcare was perfect for this project because it had a large enough data set with several different columns to analyze important predictive features like booking_date, appointment date, age, gender, waiting days, sms reminders, no shows, and previous no shows. The booking Date displays when the patient scheduled the appointment. This helps analyze booking behavior and lead time. The actual date of the scheduled appointment. It is used to understand attendance patterns over time. Age of the patient in years. It can be used to analyze how different age groups impact appointment attendance. Gender of the patient (Male/Female). Useful for demographic-based analysis. Indicates whether the patient received an SMS reminder before the appointment. This shows that if a reminder occurs, it increases the likelihood of clients attending appointments.\

Exploratory Data Analysis (EDA)

- *Target distribution*

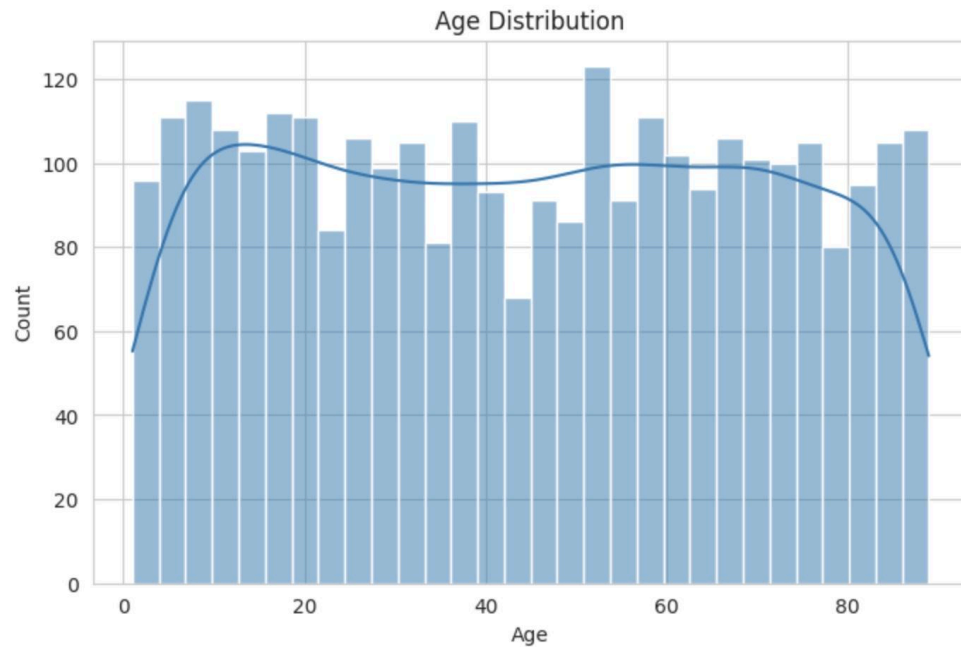


```
no_show
0      60.1
1      39.9
Name: proportion, dtype: float64
```

The target distribution shows that most patients attended their appointments, while a smaller percentage missed them. This class imbalance is important because it can affect model performance and evaluation metrics. Understanding the distribution helps guide model selection and interpretation.

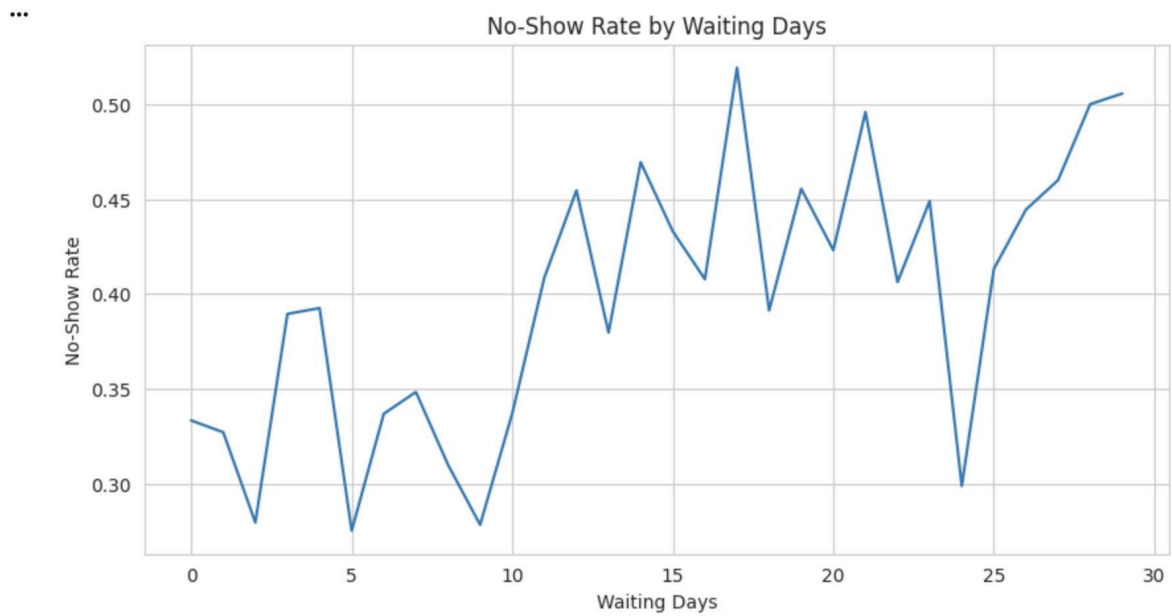
- *Age histogram*

...



The age histogram shows that the dataset contains patients from a wide range of age groups, with a higher concentration of younger and middle-aged patients. Younger patients appeared more likely to miss appointments compared to older patients. This suggests that age may be an important predictive feature in the models.

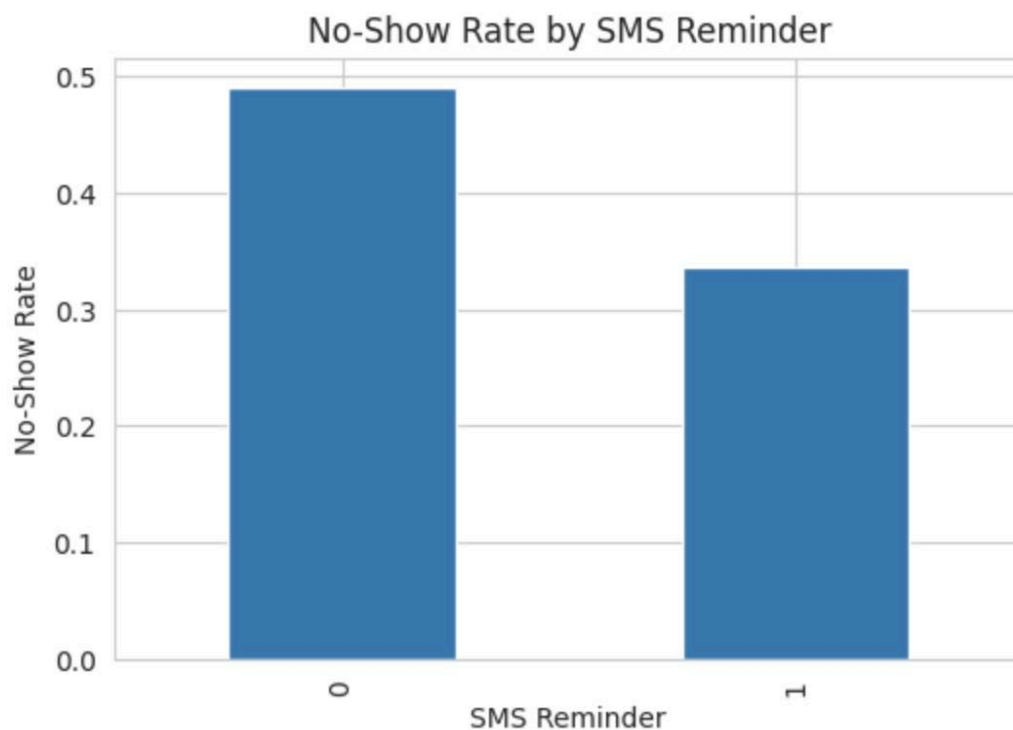
- *Waiting days plot*



The waiting days plot showed that patients with longer waiting periods were more likely to miss appointments. This pattern suggests that scheduling delays may reduce patient commitment or increase the likelihood of conflicts. Because of this relationship, `waiting_days` became one of the most important engineered features used in modeling.

- *SMS reminder plot*

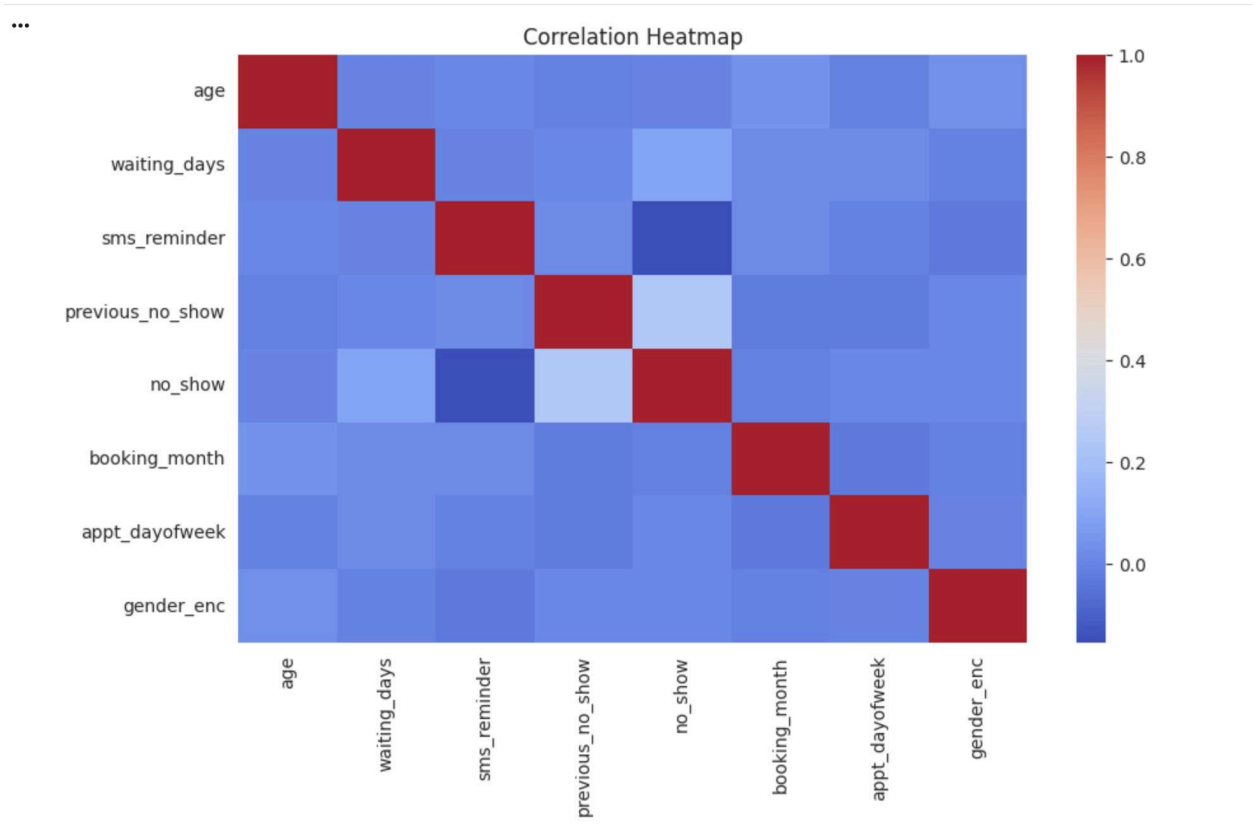
...



```
sms_reminder
0    0.491286
1    0.337047
Name: no_show, dtype: float64
```

The SMS reminder analysis showed that patients who received reminders were slightly more likely to attend their appointments. Although the effect was not as strong as features such as `waiting_days` or `previous_no_show`, reminders still appeared to positively influence attendance behavior. This suggests that reminder systems may help reduce missed appointments.

- *Correlation heatmap*



The correlation heatmap was used to examine relationships between numerical variables in the dataset. While most variables showed weak correlations, features such as `waiting_days` and `previous_no_show` demonstrated stronger relationships with the target variable. This supported their inclusion as important predictive features in the final models.

Data Cleaning & Preprocessing

- *Datetime conversion*
- *Categorical encoding*
- *Negative value correlation*

- *train/test splitting*
- *Appointment day was already included, but separating these appointments specifically was important because:*
 - *Booking_month: This was added to see if certain months people miss more often. For example, if holiday months occur and have higher no-show rates, that suggests that holidays could be a factor. Or if in the winter now shows occur more, maybe it's because clients have a hard time commuting in or don't want to risk it.*
 - *Appt_dayofweek: This is helpful to find out when and why clients are missing appointments. For example, if Saturdays and Sundays have the highest no-shows, maybe it's because people are trying to enjoy their time off from work or school.*

```
# Convert date columns
```

```
df['booking_date'] = pd.to_datetime(df['booking_date'])
```

```
df['appointment_date'] = pd.to_datetime(df['appointment_date'])
```

```
# FEATURE ENGINEERING:
```

```
# Calculate waiting days between booking and appointment
```

```
df['waiting_days'] = (
```

```
    df['appointment_date'] - df['booking_date']
```

```
).dt.days
```

```
# Fix negative waiting days

df['waiting_days'] = df['waiting_days'].apply(

    lambda x: 0 if x < 0 else x

)

# Extract month and weekday

df['booking_month'] = df['booking_date'].dt.month

df['appt_dayofweek'] = df['appointment_date'].dt.dayofweek

print("Feature engineering complete.")
```

Business Analytics Questions

Question 1

Can patient no-shows be predicted accurately using machine learning?

Yes, there may be exceptions, but when more factors are taken into account, it's possible to identify factors exacerbating the issue, but there are always outliers.

Question 2

What factors most strongly influence missed appointments?

Wait times, the longer the wait, the more likely they were to miss their appointment, previous no-shows, indicating it could be related to habits; if you've missed before, you might miss again. And younger clients are more likely to no-show here, the significance:

Previous_no_show: 0.296750

Age: 0.211629

Waiting_days: 0.177578

Question 3

Do SMS reminders reduce no-show behavior?

Yes, but not as significant as the other three mentioned, there's only a difference of 0.154239 in the number of clients showing up after being contacted via sms.

Predictive Modeling

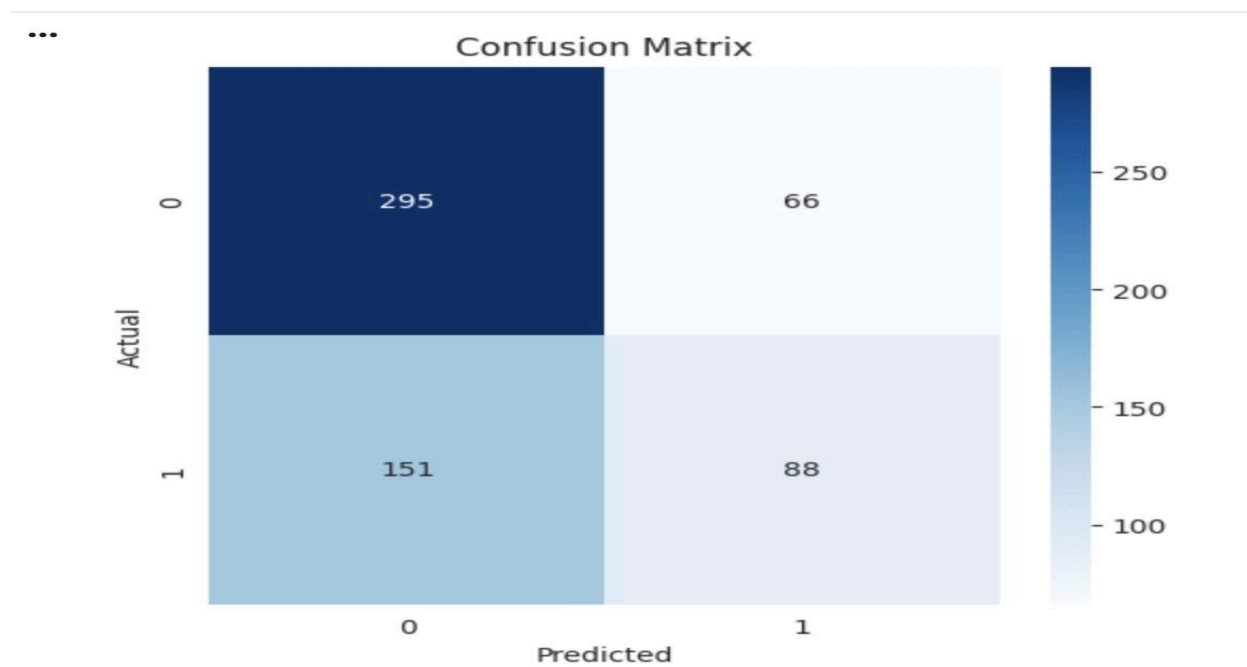
Logistic Regression was used as a baseline model, while Random Forest and Gradient Boosting were used to capture more complex relationships within the data. I also just wanted multiple options to look at to compare accuracy.

Model Evaluation & Results

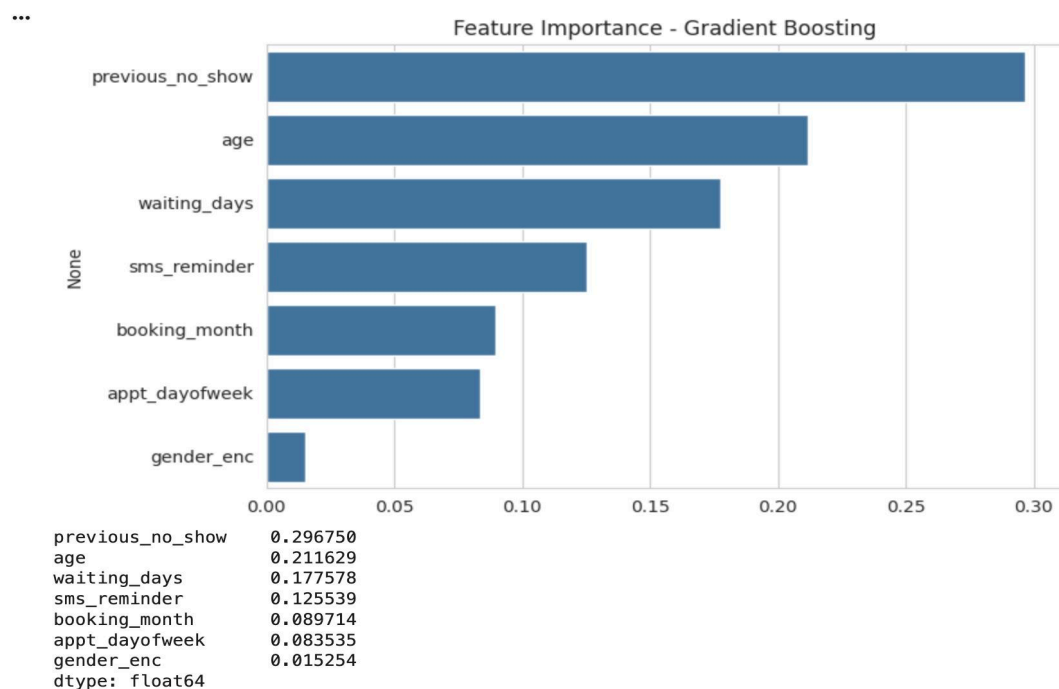
- Results table

...	Accuracy	Precision	Recall	F1 Score	ROC-AUC	CV Mean	Train Time	Inference MS
Logistic Regression	0.658333	0.626866	0.351464	0.450402	0.722945	0.650	0.045747	1.926661
Random Forest	0.625000	0.539773	0.397490	0.457831	0.641170	0.617	0.442672	20.582914
Gradient Boosting	0.638333	0.571429	0.368201	0.447837	0.694798	0.644	0.306998	2.966881

- Confusion matrix



- Feature importance



Gradient Boosting achieved the strongest overall performance and was selected as the final model for deeper analysis. It's because it was the most accurate, but also it displayed the added features better tailored to the question, versus the categories selected individually that had already been provided by the data set to get a more detailed/relevant output that covered multiple factors.

Insights & Recommendations

- *Longer waiting days increase the likelihood of a no-show*
- *Previous no-shows are likely to no-show again*
- *SMS reminders increase the likelihood of clients attending their appointment*
- *Younger patients miss appointments more than other clients*

Recommendations:

- *Targeted reminders towards clients who no-show more frequently via sms a week a day before, and the day of.*
- *Monitoring repeat no-shows, considering implementing a fee, or even terminating them as a client.*

Ethics & Interpretability

- Possible bias: This dataset is limited; a larger sample, say, 500,000 clients, may have varying results.
- Fairness concerns: It is unfair to target younger clients with more reminders just because statistically they do not show up as often.
- Healthcare ethics: Is it ethical to deny anyone service regardless of how many appointments they've missed? What if you terminate a client who cannot make appointments due to the state of their health, even if they have dementia and forget they even have an appointment? What if you miss three appointments? You're terminated. But a specific client misses one every 5 years, and after the third time (15 years later), you lose a loyal client? Should exceptions be made? And what would that look like?
- Model transparency is important to let clients know why changes are going to be made and what those changes are before holding them accountable. Maybe implement a grace period for a short time as clients learn about the new rules.

Conclusion

In conclusion, no shows are a large issue for the healthcare industry, as displayed by the dataset. It is important to conduct research to assess what predictive factors are contributing to

this significant financial loss and lack of efficiency within our Healthcare Systems. If we address the issue at hand, not only will we save money, but we will also be getting people the help they need quicker and avoid these long wait lists, so people like my coworker can be seen sooner to get the help they need for their illnesses.

Appendix

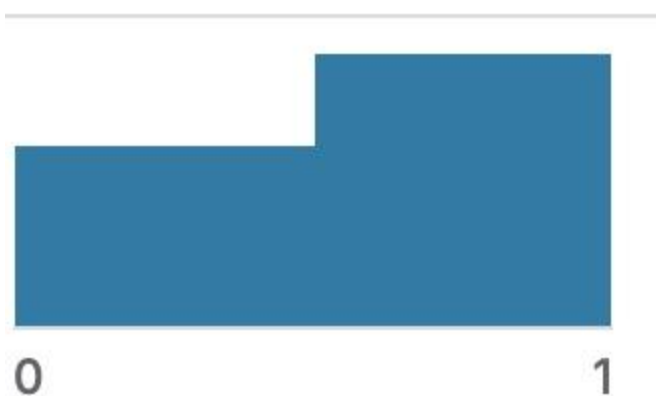
sms_reminder 

Indicates whether the patient received an SMS reminder before the appointment.



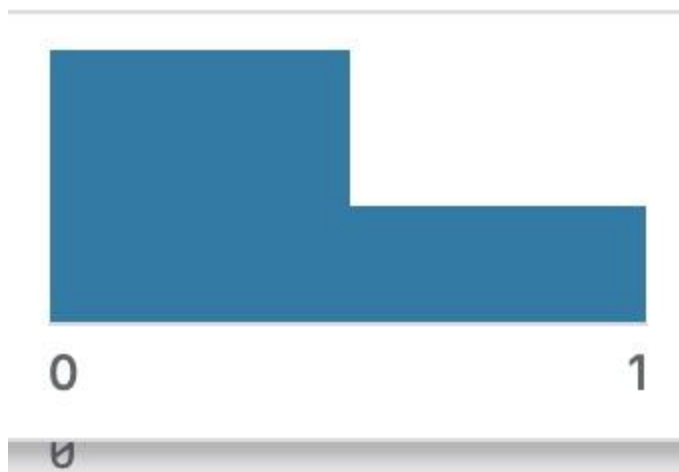
sms_reminder 

Indicates whether the patient received an SMS reminder before the appointment.



previous_no_show 

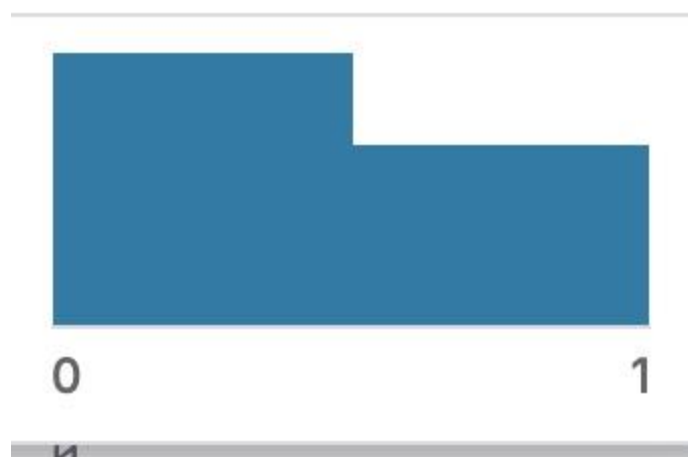
Indicates whether the patient has a history of missing appointments.



no_show



Target variable indicating whether the patient missed the appointment.



A gender

Gender of the patient (Male/Female). Useful for demographic-based analysis.

Female	50%
---------------	------------

Male	50%
-------------	------------

📅 booking_date

Date when the patient scheduled the appointment. This helps analyze booking behavior and lead time



2023-12-05 2024-12-29

2024-12-15

appointment_date

The actual date of the scheduled appointment.

Used to understand attendance patterns over time.

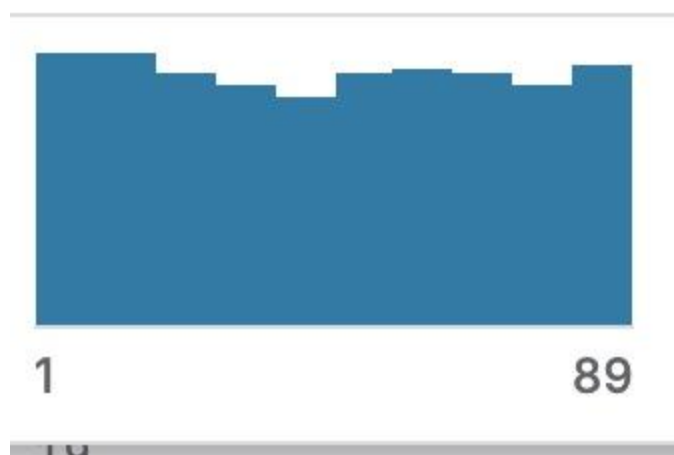


2023-12-31 2024-12-29

age



Age of the patient in years. Can be used to analyze how different age groups impact appointment attendance.



Citation

Gasida , T. (2025). *Patient No-Shows Are Costing Your Organization More than You Think.*

Here's How to Effectively Reduce Them. Artera.

<https://artera.io/blog/patient-no-shows/#:~:text=than%20You%20Think.-,Here's%20How%20to%20Effectively%20Reduce%20Them.,use%2C%20and%20delayed%20preventive%20screenin>
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